

Haoran Sun

✉ sunhaoran0301@stu.pku.edu.cn • 🌐 haoran0301.github.io • 🎓 Google Scholar

Education

Peking University

PhD Student, advised by Prof. Xiaotie Deng

Center on Frontiers of Computing Studies, School of Computer Science

Beijing

Aug 2024 – Jul 2029

Peking University

B.E. in Artificial Intelligence and B.A. in Economics (Double Major)

Tong Class, Yuanpei College

Beijing

Aug 2020 – Jul 2024

GPA: 3.7/4.0.

Selected Conference Papers (* indicates equal contribution)

- C1 RL-VLA³: Reinforcement Learning VLA Accelerating via Full Asynchronism**
Haoran Sun*, Yongjian Guo*, Zhong Guan*, Shuai Di, Xiaodong Bai, Jing Long, Tianyun Zhao, Mingxi Luo, Hongke Zhao, Likang Wu, Xiaotie Deng, Xu Chu, Xi Xiao, Sheng Wen, Yicheng Gong, Junwu Xiong
In Proceedings of the 3rd Conference on Language Modeling (COLM), 2026
- C2 Enhancing Affine Maximizer Auctions with Correlation-Aware Payment**
Haoran Sun, Xuanzhi Xia, Xu Chu, Xiaotie Deng
In Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026
- C3 Benefits and Pitfalls of Reinforcement Learning for Language Model Planning: A Theoretical Perspective**
Siwei Wang*, Yifei Shen*, **Haoran Sun***, Shi Feng*, Shang-Hua Teng, Li Dong, Yaru Hao, Wei Chen
In Proceedings of the 14th International Conference on Learning Representations (ICLR), 2026
- C4 Game Theory Meets Large Language Models: A Systematic Survey**
Haoran Sun, Yusen Wu, Yukun Cheng, Xu Chu
In Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), 2025
- C5 Scalable Virtual Valuations Combinatorial Auction Design by Combining Zeroth-Order and First-Order Optimization Methods**
Zhijian Duan*, **Haoran Sun***, Yichong Xia, Siqiang Wang, Zhilin Zhang, Chuan Yu, Jian Xu, Xiaotie Deng
In Proceedings of the ACM Web Conference 2026 (WWW)
- C6 Mechanism Design for LLM Fine-tuning with Multiple Reward Models**
Haoran Sun, Yurong Chen, Siwei Wang, Wei Chen, Xiaotie Deng
In Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS), 2025
- C7 A Scalable Neural Network for DSIC Affine Maximizer Auction Design**
Zhijian Duan, **Haoran Sun**, Yurong Chen, Xiaotie Deng
In Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023 (Spotlight)

Selected Preprints (* indicates equal contribution)

- P1 JoyNexus: Service-Oriented Multi-Tenant Post-Training for VLA Models**
Haoran Sun*, Wentao Zhang*, Junyang Hua, Hedan Yang, Yongjian Guo, Yifei Zhang, Xiaolong Xiang, Mingxi Luo, Jing Long, Chen Zhao, Chen Zhou, Wanting Xu, Qiming Yang, Hui Zhang, Song Wang, Xiaodong Bai, Shuai Di, Xu Chu, Xiaotie Deng, Yicheng Gong, Junwu Xiong
arXiv preprint, 2026

P2

LERA: LLM-Enhanced RAG for Ad Auction in Generative Chatbots
Haoran Sun, Xinrui Song, Xinyu Zhang, Zhaohua Chen, Xu Chu, Zhilin Zhang, Chuan Yu, Jian Xu, Bo Zheng, Xiaotie Deng
Presented at the NExT-Game Workshop at the 43rd International Conference on Machine Learning (ICML), 2026

P3

NoiseGate: Learning Per-Latent Timestep Schedules as Information Gating in World Action Models
Wen Huang*, **Haoran Sun***, Yongjian Guo*, Yunxuan Ma, Haoran Li, Jing Long, Zhouying Mo, Zhong Guan, Yucheng Guo, Shuai Di, Junwu Xiong
Presented at the RLxF Workshop at the 43rd International Conference on Machine Learning (ICML), 2026

P4

Missing Old Logits in Asynchronous Agentic RL: Semantic Mismatch and Repair Methods for Off-Policy Correction
Zhong Guan*, Yongjian Guo*, **Haoran Sun***, Wen Huang, Shuai Di, Junwu Xiong, Likang Wu, Hongke Zhao
Presented at the FAGEN Workshop at the 43rd International Conference on Machine Learning (ICML), 2026

P5

Thousand-GPU Large-Scale Training and Optimization Recipe for AI-Native Cloud Embodied Intelligence Infrastructure
JDT AI Infra Team
arXiv preprint, 2026

Internships

AI Infra Team, JD Technology, JD	Sept 2025 – Now
Engineering efficient and high-performance training algorithms for VLA models, mentored by Junwu Xiong.	
Advertising Group, Alimama, Alibaba Group	Feb 2025 – Sept 2025
Investigating mechanism design problems for online advertising, mentored by Zhilin Zhang.	
Theory Group, Microsoft Research Asia	Jun 2023 – Mar 2024
Researched the foundations of reasoning in LLMs under the mentorship of Wei Chen.	

Academic Service

Workshop Organizer	
ICLR 2026 Workshop on AI for Mechanism Design and Strategic Decision Making (AIMS).	
Peer Review	
- Conferences: ICLR 2026, NeurIPS (2024-2026), ICML (2024-2026), WWW 2024, AAAI 2026, COLM 2026. - Journal: IEEE Transactions on Consumer Electronics (TCE).	

Selected Awards and Honors

Peking University President Scholarship for PhD Students	June 2026
Peking University Song Qingling Scholarship	June 2024
Peking University Yang Fuqing & Wang Yangyuan Academician Scholarship	Oct 2023
NeurIPS Auto-Bidding in Large-Scale Auctions Competition	Dec 2024
3rd Place in General Track, NeurIPS 2024.	

Skills, Interests & Others

Programming Languages:

Python, $\LaTeX$, C/C++

Frameworks & Tools:

PyTorch, Transformers, SGLang

Languages:

Chinese (Native), English (CET-6: 560, CET-4: 622)